Multi-Client Quiz System
Report

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1 Protocol design

The quiz system uses a binary header + JSON payload protocol over TCP sockets. Every message begins with a fixed-size header and an optional JSON payload.

1.1 Message structure

- Header: `struct MessageHeader { MessageType type; uint32_t payload_size; }` (packed).
- Payload: UTF-8 encoded JSON string or plain string depending on message type.
- Byte order: integers sent in network byte order using `htonl/ntohl`.

1.2 Client to Server Messages

- `MSG_GENRE` – selects quiz genre (string).
- `MSG_ANSWER` – answer JSON { "answer":"A" }.
- `MSG_LEADERBOARD_REQ` – request leaderboard.
- `MSG_ALIVE_OK` – heartbeat response.

1.3 Server to Client Messages

- `MSG_QUESTION` – JSON { "q": question, "options": [...], ("cached":true/false) }.
- `MSG_FEEDBACK` – JSON { "result":"CORRECT"/"WRONG", "score": int }.
- `MSG_LEADERBOARD_RESP` – JSON array of players and scores.
- `MSG_QUIZ_OVER` – final score string.
- `MSG_KEEP_ALIVE` – probe message.
- `MSG_ERROR` – error string.

2 Testing evidence for 1A

This section contains console logs and screenshots for the Part 1A system (`server.cpp`, `client.cpp`).

Expected evidence

- Server log: client connects, requests genre, questions fetched from LLM.
- Client log: displays questions, sends answers, receives feedback and final score.
- Leaderboard requests and responses.
- Keep-alive probe and response.

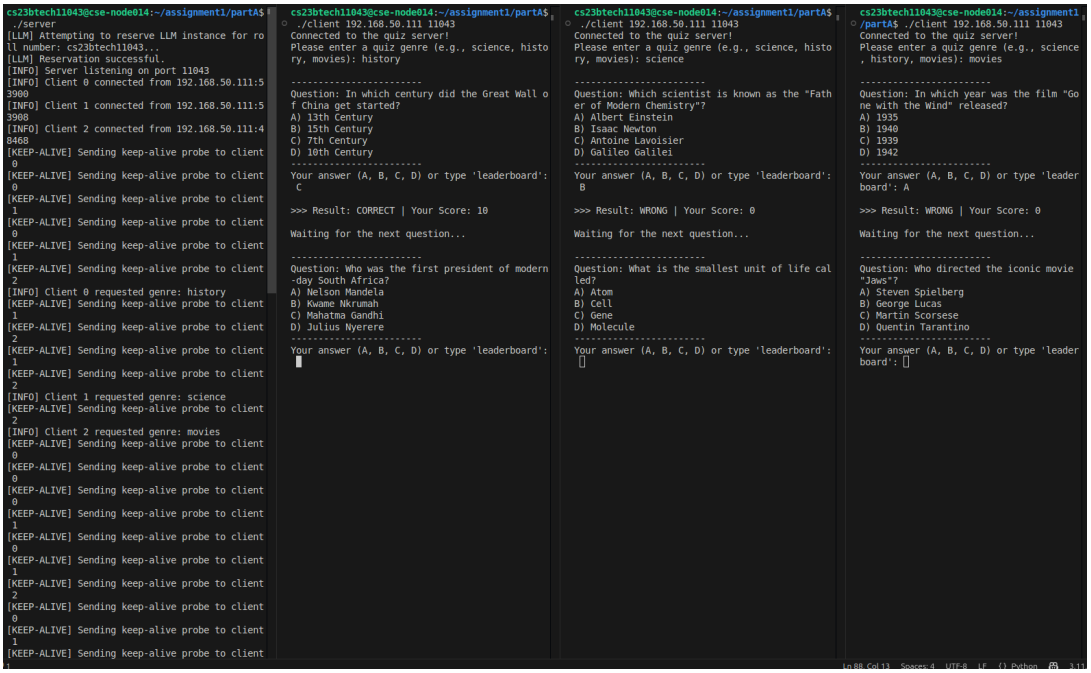


Figure 1: Example evidence for Part 1A (client-server exchange without leaderboard).

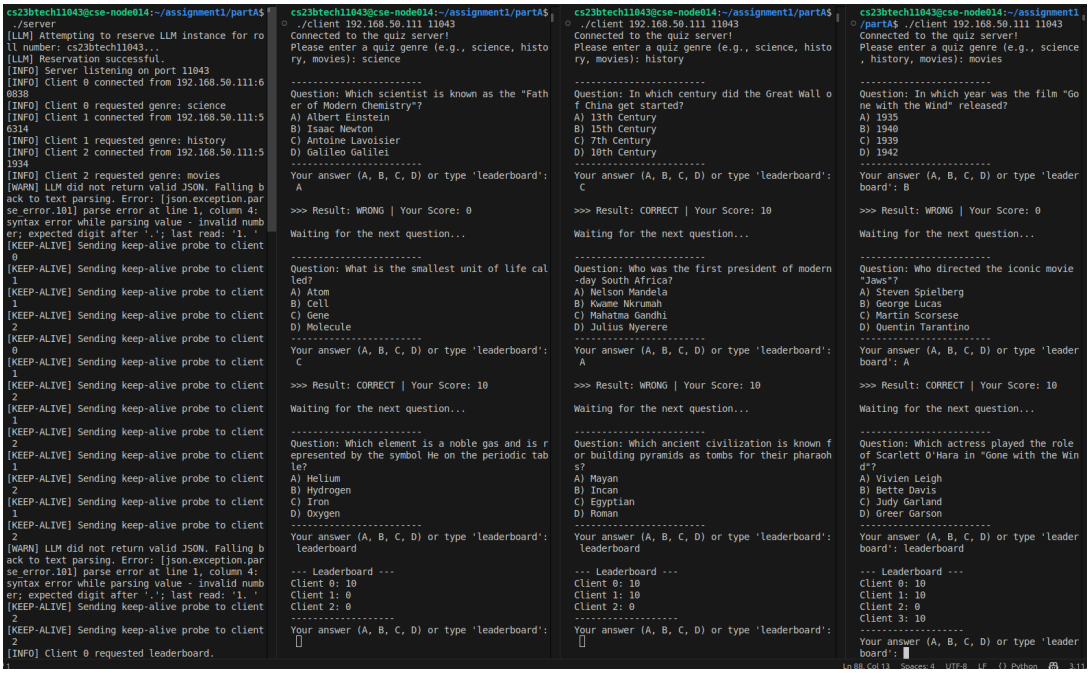


Figure 2: Example evidence for Part 1A (client-server exchangew with leaderboard).

3 Testing evidence for 1B

This section contains console logs and screenshots for the Part 1B system (`server_B.cpp`, `client_B.cpp`).

Expected evidence

- Server log: cache miss for first genre request, cache hit for later requests.

- Client log: questions displayed with a "(Question from cache)" or "(Freshly generated question)" label.
- Cache expiration messages after timeout.

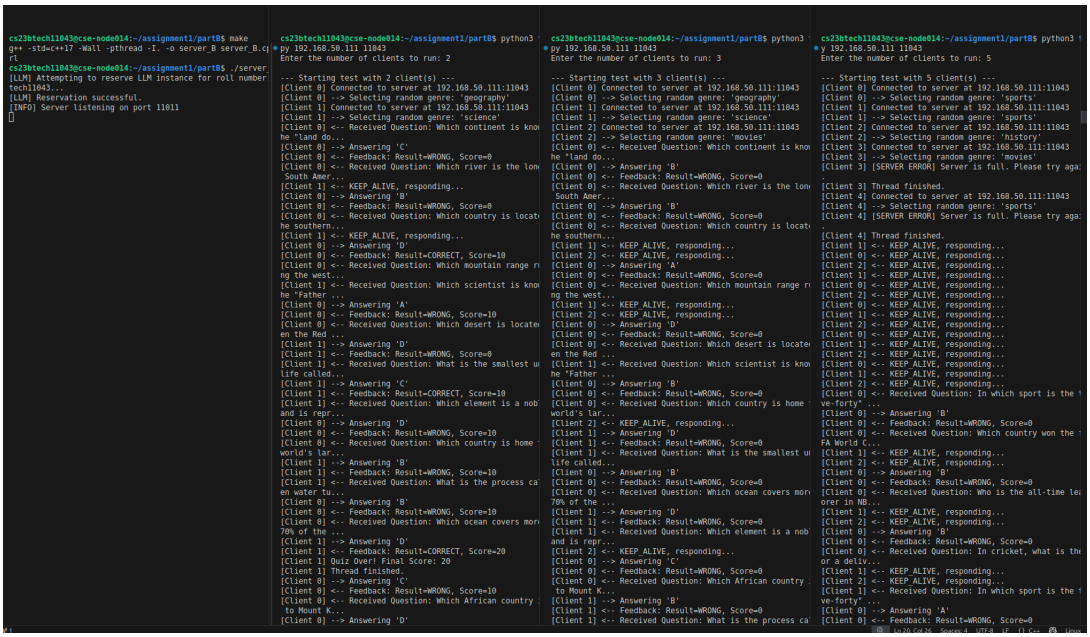


Figure 3: Example evidence for Part 1B (cache functionality visible).

4 Performance Measurement and Traffic Analysis

Performance was tested with `tester.py` by running multiple clients concurrently. Metrics: initial question latency, answer latency, throughput per client.

4.1 Performance Graphs

4.1.1 Latency (Waiting Time)

- Part A: latency grows with number of clients (LLM bottleneck).
- Part B: cache reduces initial question latency for repeated genres.

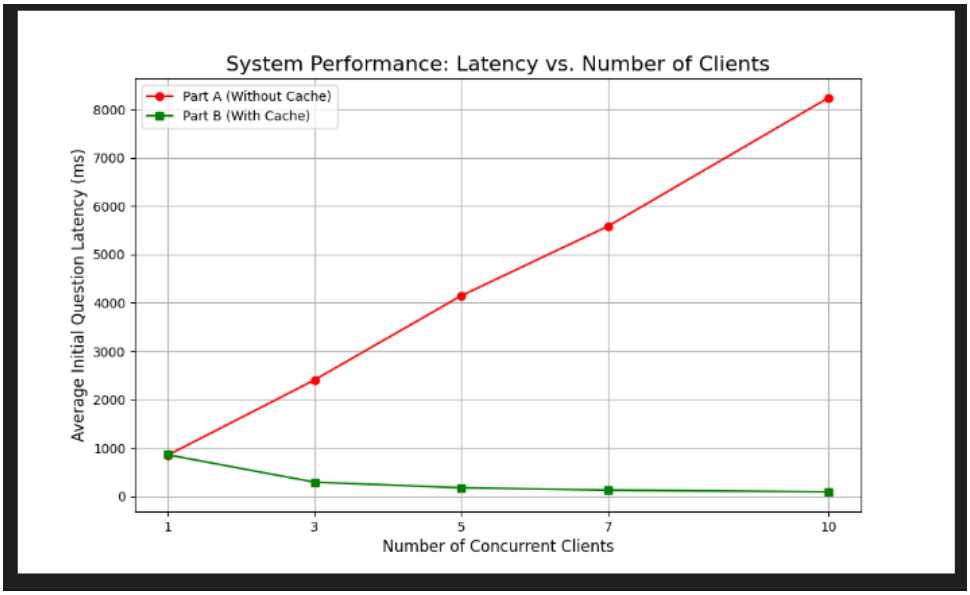


Figure 4: Latency comparison between Part A and Part B.

4.1.2 Throughput (How Much Stuff Gets Done)

- Throughput per client decreases as clients increase.
- Part B maintains higher throughput due to caching.

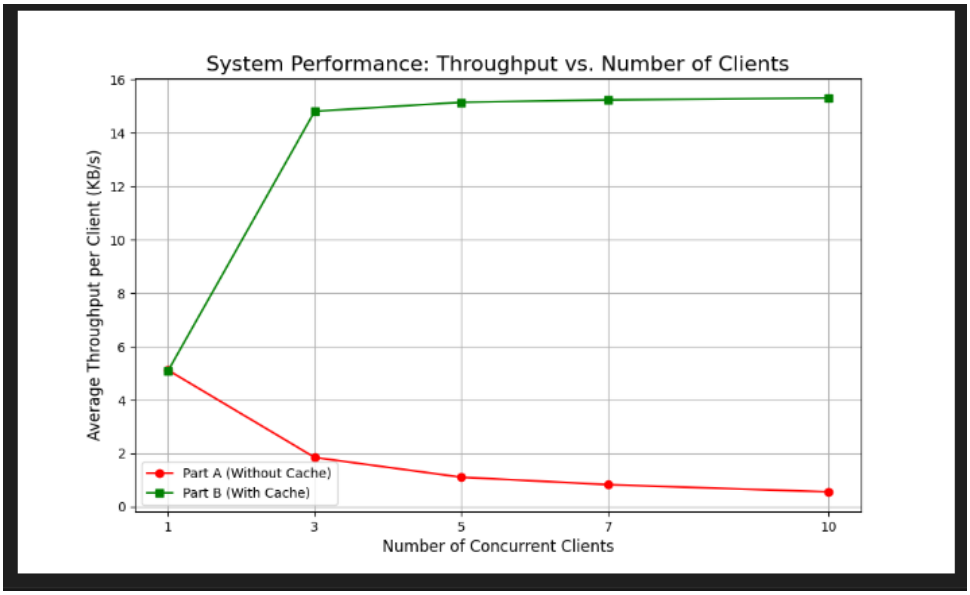


Figure 5: Throughput per client for different concurrency levels.

4.2 Bottleneck Identification

- Part A: LLM request/response time is the bottleneck; question generation cost dominates.
- Part B: Cache mitigates LLM bottleneck, so network bandwidth and server CPU become more visible constraints.
- For very high concurrency, throughput is bounded by server's socket I/O and JSON parsing speed.

5 PLAGIARISM STATEMENT

We certify that this assignment/report is our own work based on our personal study and/or research and that we have acknowledged all material and sources used in its preparation, whether they be books, articles, packages, datasets, reports, lecture notes, and any other kind of document, electronic or personal communication. We also certify that this assignment/report has not previously been submitted for assessment/project in any other course lab, except where specific permission has been granted from all course instructors involved, or at any other time in this course, and that we have not copied in part or whole or otherwise plagiarized the work of other students and/or persons. We pledge to uphold the principles of honesty and responsibility at CSE@IITH. In addition, We understand my responsibility to report honor violations by other students if we become aware of it.

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